

Full production three-component gradient flow: integrated referee theorem

TECT verification-first repository · claim A2-FULL-PRODUCTION-WELLPOSED

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1. Purpose and scope

This document is the self-contained consolidation of the v1.0–v1.3 proof sequence. The operator independently executed the four audit programs and on 2026-07-17 instructed the repository to determine and enact the justified tier. The resulting classification is T6 CONDITIONAL-THEOREM, conditional only on the named definitional input **A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL**.

The mathematical domain is a fixed three-torus $\mathbb{T}^3 = [0, 16]^3$, with field $\Psi : \mathbb{T}^3 \rightarrow \mathbb{C}^3$ regarded as six real components. The pairing is

$$\langle U, V \rangle_{\mathbb{R}} = \operatorname{Re} \int_{\mathbb{T}^3} U^\dagger V \, dx.$$

The functional is exactly the hash-pinned P1 reference functional, with the pinned production coefficients, positive rho and Class-II mass regularisers, and $\eta_{\text{shell}} = 0$. The historical solver backend is a non-variational proxy and is excluded. Also excluded are nonzero shell bias, initial data below H^2 , infinite volume, negative shell mass, minimiser selection, BCC selection, and stability of any particular vacuum.

2. Functional and theorem

Let $\rho = \Psi^\dagger \Psi$, let T_A be the three P1 Hermitian generators, and define

$$m_A = \Psi^\dagger T_A \Psi, \quad J_A = \nabla m_A, \quad K_A = J_A - \frac{m_A}{\rho + \epsilon_\rho} \nabla \rho, \quad \epsilon_\rho = 10^{-12}.$$

The functional is

$$\begin{aligned} \mathcal{F}(\Psi) = & \frac{1}{2} \langle \Psi, L \Psi \rangle_{\mathbb{R}} + \int_{\mathbb{T}^3} \left(\frac{\lambda}{4} \rho^2 + \frac{\gamma}{6} \rho^3 \right) dx \\ & + \sum_A \int_{\mathbb{T}^3} \left(\frac{a}{2} |J_A|^2 + b \operatorname{Re}(J_A^\dagger K_A) + \frac{c}{2} |K_A|^2 \right) dx, \end{aligned} \quad (2.1)$$

where

$$L = (r + Z(-\Delta) + Y(-\Delta)^2) I_3 + M_{\text{fam}} + k_{\text{lock}}(I_3 - P_0), \quad (2.2)$$

and

$$a = \frac{c_{JJ} \alpha_X^2}{M_X^2 + \epsilon_M}, \quad b = \frac{c_{JK} \alpha_X \beta_X}{M_X^2 + \epsilon_M}, \quad c = \frac{c_{KK} \beta_X^2}{M_X^2 + \epsilon_M}, \quad \epsilon_M = 10^{-12}. \quad (2.3)$$

Theorem. Assume **A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL**. For every $\Psi_0 \in H^2(\mathbb{T}^3; \mathbb{C}^3)$, the real- L^2 gradient flow

$$\partial_t \Psi = -\nabla_{\mathbb{R}} \mathcal{F}(\Psi), \quad \Psi(0) = \Psi_0, \quad (2.4)$$

has a unique global solution

$$\Psi \in C([0, \infty); H^2) \cap L_{\text{loc}}^2((0, \infty); H^4), \quad \partial_t \Psi \in L_{\text{loc}}^2((0, \infty); L^2). \quad (2.5)$$

On every finite interval the solution depends continuously on the initial datum in H^2 . It satisfies the exact energy identity

$$\mathcal{F}(\Psi(t)) + \int_s^t \|\partial_\tau \Psi(\tau)\|_{L^2}^2 d\tau = \mathcal{F}(\Psi(s)), \quad 0 \leq s \leq t, \quad (2.6)$$

and is C^∞ in space and time for every $t > 0$.

3. Linear operator and coercivity

The scalar Fourier symbol is $p(x) = r + Zx + Yx^2$ with $x = |k|^2$, $Y = 1$, $Z = -0.9252754126$, and $r = 0.4740336473$. Completing the square gives

$$\min_{x \geq 0} p(x) = r - \frac{Z^2}{4Y} = 0.260000000009475 > 0. \quad (3.1)$$

Moreover

$$\inf_{x \geq 0} \frac{p(x)}{1 + x^2} = 0.2048572626782363 > 0. \quad (3.2)$$

The family matrix is positive semidefinite, while P_0 is an orthogonal projector and $k_{\text{lock}}(I - P_0)$ is positive semidefinite. Thus L is positive self-adjoint on H^4 , its form domain is H^2 , and $D(L^{1/2}) = H^2$ with equivalent norms.

The Class-II coefficient matrix is

$$Q_{II} = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad \det Q_{II} = 7.031249999996483 \times 10^{-6}, \quad \lambda_{\min}(Q_{II}) = 0.001259011500926061 > 0. \quad (3.3)$$

Hence the Class-II energy is nonnegative. Finally $\lambda = -0.43$ and $\gamma = 1.62 > 0$, and Young's inequality gives

$$\frac{|\lambda|}{4} \rho^2 \leq \frac{\gamma}{12} \rho^3 + C_Y, \quad C_Y = 0.010098435197886498. \quad (3.4)$$

Consequently

$$\mathcal{F}(\Psi) \geq \frac{c_{H^2}}{2} \|\Psi\|_{H^2}^2 + \frac{\gamma}{12} \|\Psi\|_{L^6}^6 - C_Y |\mathbb{T}^3|. \quad (3.5)$$

4. Full nonlinear map and local solution

Realify Ψ as $u \in \mathbb{R}^6$. For each generator let A_A be its real symmetric realification and define

$$p_A(u) = 2A_A u, \quad q_A(u) = \frac{u^T A_A u}{u^T u + \epsilon_\rho}, \quad s_A(u) = 2(A_A - q_A(u)I)u. \quad (4.1)$$

The Class-II density is exactly

$$G_{II}(u, \nabla u) = \frac{1}{2} \sum_j (\partial_j u)^T B(u) (\partial_j u), \quad (4.2)$$

where

$$B(u) = \sum_A \{a p_A p_A^T + b(p_A s_A^T + s_A p_A^T) + c s_A s_A^T\}. \quad (4.3)$$

Because $\epsilon_\rho > 0$, B is smooth on all of $\mathbb{R}^6$. Direct Euler–Lagrange differentiation gives only

$$N_{II}(u) = B(u)\nabla^2 u + C(u)[\nabla u, \nabla u], \quad (4.4)$$

with smooth C ; no derivative above order two occurs. On a fixed three-torus, $H^2 \hookrightarrow L^\infty \cap W^{1,6}$ and $L^3 \hookrightarrow L^2$. Product and difference estimates therefore yield

$$\|N(u) - N(v)\|_{L^2} \leq C_R \|u - v\|_{H^2} \quad \text{on every bounded } H^2 \text{ ball.} \quad (4.5)$$

The quartic and sextic terms obey the same bound because H^2 is an algebra.

The positive operator L generates an analytic semigroup and

$$\|L^{1/2}e^{-tL}\|_{L^2 \rightarrow L^2} \leq Ct^{-1/2}. \quad (4.6)$$

The kernel is integrable. The mild equation and (4.5)–(4.6) give a contraction in $C([0, T]; H^2)$, hence a unique local solution and the continuation alternative that finite maximal time requires the H^2 norm to diverge.

5. Energy identity and global continuation

Let P_n be the real L^2 -orthogonal Fourier projection and solve

$$\partial_t u_n = -P_n(Lu_n + N(u_n)). \quad (5.1)$$

This is the exact finite-dimensional real gradient flow of $\mathcal{F}|_{P_n H}$, so

$$\frac{d}{dt} \mathcal{F}(u_n) = -\|\partial_t u_n\|_{L^2}^2. \quad (5.2)$$

The coercive bound gives u_n uniformly in $L^\infty(0, T; H^2)$, while (5.2) bounds $\partial_t u_n$ in $L^2(0, T; L^2)$. Since $Lu_n = -\partial_t u_n - P_n N(u_n)$, ellipticity also bounds u_n in $L^2(0, T; H^4)$. Aubin–Lions gives strong convergence in $L^2(0, T; H^2)$; (4.5) gives strong convergence of $N(u_n)$ in $L^2(0, T; L^2)$.

For the nonlinear potential Φ , the identity $D\Phi(u)[h] = \langle N(u), h \rangle_{\mathbb{R}}$ and the $H^2 \rightarrow L^2$ local Lipschitz bound give the chain rule

$$\Phi(u(t)) - \Phi(u(s)) = \int_s^t \langle N(u), \partial_t u \rangle_{\mathbb{R}} dt. \quad (5.3)$$

Combining (5.3) with the spectral quadratic-form chain rule proves (2.6), not merely an inequality. Equations (2.6) and (3.5) prevent the H^2 continuation alternative, proving global existence.

6. Continuous dependence and positive-time smoothing

For two solutions u, v in a common energy-bounded H^2 ball, the mild difference equation gives

$$\|u(t) - v(t)\|_{H^2} \leq C\|u_0 - v_0\|_{H^2} + C_R \int_0^t (t-s)^{-1/2} \|u(s) - v(s)\|_{H^2} ds. \quad (6.1)$$

Weakly singular Gronwall proves continuous dependence uniformly on every finite interval.

For each $\alpha < 1$, analytic-semigroup estimates first give $u(t) \in D(L^\alpha) = H^{4\alpha}$ for $t > 0$. Choosing $\alpha > 1/2$ makes u Hölder continuous in H^2 away from zero; then $f(t) = N(u(t))$ is Hölder continuous into L^2 . The endpoint H^4 gain follows from the cancellation

$$L \int_a^t e^{-(t-s)L} f(s) ds = (I - e^{-(t-a)L})f(t) + \int_a^t L e^{-(t-s)L} [f(s) - f(t)] ds. \quad (6.2)$$

The last integrand is bounded by $C(t-s)^{\theta-1}$ for some $\theta > 0$ and is therefore integrable. Hence $u(t) \in D(L) = H^4$ for every $t > 0$.

The positive regularisers make all coefficient functions smooth, and the nonlinear map has differential order two. Periodic Moser estimates give

$$N : H^m \rightarrow H^{m-2} \quad \text{locally Lipschitz for every integer } m \geq 2. \quad (6.3)$$

Repeating the endpoint argument on the base space H^{m-2} gains two derivatives at each positive-time step. Thus $u(t)$ lies in every Sobolev space for $t > 0$; the equation then supplies all time derivatives. This proves the stated C^∞ regularity.

7. Independent verification matrix

Four standalone NumPy audits form the executable verification matrix:

audit	assertions	load-bearing maximum or minimum
coercivity baseline	20/20	$c_{H^2} = 0.2048572626782363$
nonlinear real-coordinate map	14/14	$\text{relerr}(E_{II}) = 9.04 \times 10^{-11}$
Galerkin energy/continuation	12/12	$\text{relerr}(dF/dt) = 7.99 \times 10^{-13}$
semigroup/smoothing	15/15	fractional-bound ratio = 0.999999998916

All scripts record source hashes, environment, assertions, and JSON evidence. The operator independently reproduced the completed matrix before authorising tier determination. The programs audit signs, factors, real-pairing conventions, spectral constants, and limiting identities; Sections 3–6 are the proof.

8. Tier decision in plain terms

T5 would say only that a pinned calculation is closed. The present result is stronger: it establishes a theorem about every H^2 initial field in the declared periodic domain, rather than a finite list of runs. T6 is therefore the correct tier. It remains conditional because the physical identification of the hash-pinned P1 reference functional as the production functional is a T5 definitional input. T7 is excluded because the theorem does not cover the full physical domain: it omits the historical backend, nonzero shell bias, infinite volume, negative shell mass, and vacuum/BCC selection, and it has not undergone the T7 dual external-domain audit.

In plain language, the theorem says: once the pinned three-component energy rule is accepted, its time evolution is mathematically safe. A valid starting field cannot split into two different futures or blow up in finite time; nearby starting fields stay nearby over finite times; the energy can only decrease; and every initially rough H^2 field becomes smooth immediately after positive time. It does not say which vacuum Nature chooses or that the universe must form a BCC phase.

9. Devil’s-advocate

- “*The discrete P1 backend already proves a continuum PDE theorem.*” UPHELD as false. It is retained as the named definitional hypothesis; Sections 3–6 provide the separate continuum proof.
- “*The Class-II cross term destroys positivity.*” DISMISSED in the pinned scope by the positive determinant and minimum eigenvalue in (3.3).
- “*The rational Class-II coefficient is singular at the zero field.*” DISMISSED only because the pinned $\epsilon_\rho > 0$ makes the coefficient globally smooth. Removing the floor requires a new theorem.

- “*A finite Galerkin energy equality automatically survives the limit.*” UPHELD as an invalid shortcut. The H^4/H^2 compactness and explicit nonlinear chain rule in Section 5 are required.
- “*Fractional smoothing for every $\alpha < 1$ already gives H^4 .*” UPHELD as false. The endpoint cancellation (6.2) and positive Holder exponent are essential.
- “*This theorem proves vacuum or BCC selection.*” UPHELD as an overclaim. Well-posed dynamics does not identify a global minimiser or a preferred structure.
- “*The evidence justifies T7.*” UPHELD as a governance error. The declared physical domain is intentionally narrower and the T7 external-domain requirements are not met.

10. Result footer

Result ID:	A2-FULL-PRODUCTION-WELLPOSED
Precise statement:	The canonical full-production real-L2 gradient flow is uniquely globally well-posed in H2, continuously dependent, energy decreasing, and smooth for $t > 0$.
Scope:	Fixed T3, C3 field as six real components, hash-pinned P1 functional, positive regularisers, $\eta_{\text{shell}}=0$.
Dependencies:	A1-PRODUCTION-FUNCTIONAL-REALISATION; A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL; integrated proof and four independent executable audits.
Reproduction command:	Run the four codes/foundations/a2_full_production_*.py audit scripts in the bundle README order.
Expected output:	20/20, 14/14, 12/12, and 15/15 PASS; all four named A2-FULL verdicts PASS; every command exits 0.
Falsification gate:	Counterexample to existence, uniqueness, finite-time H2 continuation, continuous dependence, energy identity, endpoint H4 gain, or smooth bootstrap.
Tier before / after:	T4 / T6 CONDITIONAL-THEOREM, operator-confirmed; registry enactment follows final bundle integrity.
No-overclaim statement:	Not historical backend, $\eta_{\text{shell}} \neq 0$, below H2, infinite volume, negative shell mass, minimiser, BCC selection, stability, T7, or TOE closure.
Next required action:	Build the content-addressed T6 reproduction bundle, run release_check, and enact the atomic card promotion.